

Building GPT-2 from Scratch: Backbone, Optimizer, and Parameter-Efficient Downstream Adaptation

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Abstract. We reimplement the GPT-2 language model and the AdamW optimizer ourselves, load the official pre-trained weights into our own modules, and validate the result against reference implementations. Our model reproduces the HuggingFace GPT-2 hidden states to an absolute tolerance of 10^{-1} , and our optimizer step matches a reference tensor to 10^{-6} . We then adapt the model to three downstream tasks (five-way sentiment classification, Quora paraphrase detection, and Shakespearean sonnet generation) and study parameter-efficient fine-tuning (PEFT) in a controlled setting. Our central contribution is an ablation of Low-Rank Adaptation (LoRA) on the attention query/key/value projections under three initialization schemes. Training only 0.44M parameters (0.36% of the model), LoRA with a weight-distribution initialization reaches a chrF of 40.09 on held-out sonnets, which is 97.9% of the full fine-tuning score (40.95) while updating $281\times$ fewer parameters. We run the same ablation on paraphrase detection, where the three initializations instead finish within 0.1 accuracy point of one another, indicating that the choice of initialization matters for low-resource generation but not for this large-data classification task. Throughout, we report quantitative results, learning curves, the trade-off between trainable parameters and quality, and a qualitative analysis of generation behavior.

1 Introduction

The pretrain-then-finetune paradigm popularized by GPT-1 and GPT-2 underlies contemporary language modeling: a decoder-only Transformer is trained on next-token prediction over large corpora and subsequently specialized to downstream tasks [8, 9]. This project reconstructs that pipeline end to end. Rather than calling a library model, we implement the GPT-2 architecture and the AdamW optimizer ourselves, transfer the official pretrained weights into our modules, and verify numerical equivalence with reference implementations before any task-specific training.

The work has two parts. The first establishes a correct backbone (masked multi-head self-attention, the pre-LayerNorm Transformer block, learned token and position embeddings, and a weight-tied language-modeling head) together with the optimizer that trains it. The second adapts the validated model to three tasks of increasing generative complexity and studies how to adapt it *efficiently*.

Our contributions are:

- A from-scratch GPT-2 and AdamW implementation, numerically validated against the HuggingFace model (hidden-state tolerance 10^{-1}) and a reference optimizer step (tolerance 10^{-6}).
- Three downstream adaptations with quantitative dev-set results: sentiment classification on SST and CFIMDB, paraphrase detection on Quora, and conditioned sonnet generation evaluated with chrF and BLEU.
- A controlled **LoRA ablation** on the attention projections, comparing three initialization schemes against full fine-tuning and last-layer fine-tuning across both a generation task (sonnets) and a classification task (paraphrase). Weight-distribution initialization retains 97.9% of full fine-tuning chrF at 0.36% of the trainable parameters on sonnets, while on paraphrase the three initializations are essentially tied, each recovering about 94% of full accuracy at the same budget.

All reported numbers are measured on the training and development splits only; test labels are never used locally, in accordance with the dataset usage policy.

2 Background

Decoder-only Transformer. GPT-2 stacks identical Transformer blocks [11] that process tokens left to right. Each block applies masked (causal) multi-head self-attention followed by a position-wise feed-forward network, with residual connections around both sublayers. GPT-2 places layer normalization *before* each sublayer (pre-LN), which stabilizes gradient flow and removes the need for a learning-rate warmup that the original post-LN design required [1]. The output projection reuses (*ties*) the input token-embedding matrix.

AdamW. Adam maintains exponential moving averages of the gradient and its square and applies bias correction [4]; AdamW decouples weight decay from the gradient update, applying it as a direct multiplicative shrinkage of the parameters [5].

Parameter-efficient fine-tuning. LoRA freezes the pretrained weights and learns a low-rank additive update $\Delta W = (\alpha/r)BA$ for selected projections, drastically reducing the number of trainable and stored parameters [3].

Generation metrics. chrF is a character n -gram F-score suited to morphology- and form-sensitive text [7]; BLEU is a word n -gram precision metric with a brevity penalty [6]. We treat chrF as primary and BLEU as complementary, following our task definition.

3 Methodology

Figure 1 summarizes one pre-LN block, the LoRA-augmented attention projections, and the weight-tied head. We describe each implemented component below.

3.1 GPT-2 backbone

Masked multi-head self-attention. For each head we compute scaled dot-product attention

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M_{\text{causal}} + M_{\text{pad}}\right)V, \quad (1)$$

with head dimension $d_k = 64$. The causal mask M_{causal} is built from a lower-triangular matrix (`torch.tril`); positions corresponding to future tokens are set to -10^4 before

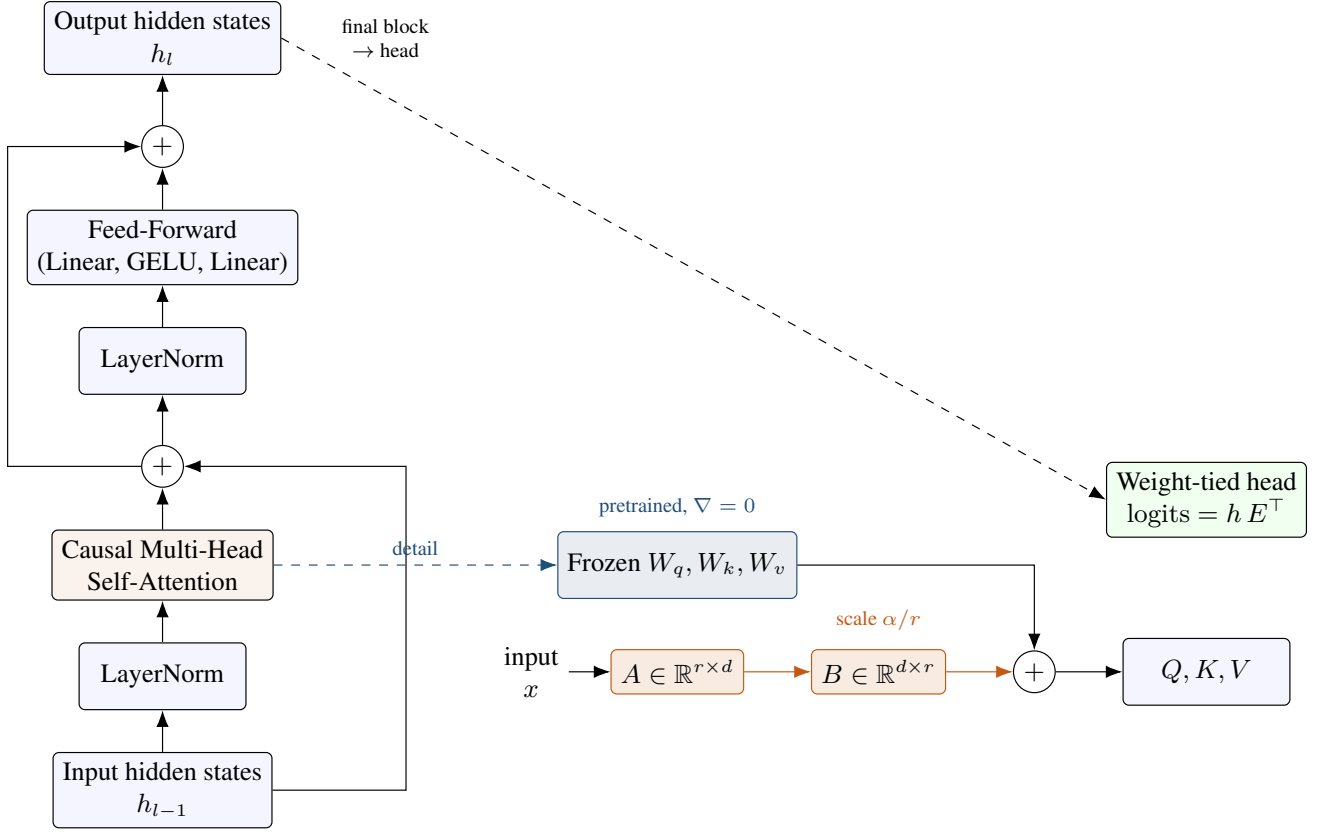


Figure 1: One pre-LayerNorm GPT-2 block (left, bottom-to-top), the LoRA-augmented attention projection (right): each frozen weight W is summed with a low-rank path $(\alpha/r)BA$ that is the *only* trainable component in LoRA mode; and the weight-tied output head used for language modeling.

the softmax so their post-softmax weight is approximately zero. The padding mask M_{pad} , derived from the attention mask, is added in the same additive form, so variable-length sequences batch cleanly. We use the finite value -10^4 rather than $-\infty$ for numerical stability.

Pre-LayerNorm block. Each block applies normalization before the sublayer and adds the residual afterward:

$$\tilde{h} = h + \text{Drop}(W_O \text{Attn}(\text{LN}_1(h))), \quad (2)$$

$$h' = \tilde{h} + \text{Drop}(W_2 \text{GELU}(W_1 \text{LN}_2(\tilde{h}))). \quad (3)$$

The dense-projection-plus-dropout-plus-residual step deliberately does not re-apply normalization; the GELU activation [2] follows GPT-2.

Embeddings and weight tying. Input representations are the sum of learned token and positional embeddings followed by dropout. Output logits reuse the transposed token-embedding matrix, $\text{logits} = h E^\top$ with $E \in \mathbb{R}^{V \times d}$, which avoids a separate $V \times d$ output matrix; for $V = 50,257$ and $d = 768$ this ties $\approx 38.6\text{M}$ parameters.

Weight transfer. We load the official GPT-2 weights into our modules [12]. The principal remapping concerns attention: HuggingFace stores Q, K, V as one fused 1D-convolution weight, which we slice into three separate linear projections and transpose to match the linear-layer convention.

3.2 AdamW optimizer

Our step uses the efficient bias-correction form. With first and second moments

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (4)$$

we fold both bias corrections into a single step size

$$\alpha_t = \alpha \frac{\sqrt{1 - \beta_2^t}}{1 - \beta_1^t}, \quad \theta_t = \theta_{t-1} - \alpha_t \frac{m_t}{\sqrt{v_t} + \epsilon}, \quad (5)$$

and then apply decoupled weight decay as a multiplicative shrinkage $\theta_t \leftarrow \theta_t (1 - \alpha \lambda)$ after the gradient step [5].

3.3 Task heads

Sentiment. The last non-padding token representation is passed through dropout and a linear classifier to the label space. Two regimes are compared: *last-linear-layer* freezes the backbone and trains only the head, while *full-model* trains all parameters.

Paraphrase. We use the last-token representation with a two-way linear head trained with cross-entropy against the binary duplicate label. We note this is a design choice: the assignment also describes a cloze-style formulation that generates the words “yes”/“no”. A practical consequence of our setup is that the train and test prompt templates differ, which we revisit in the analysis.

Sonnet generation. We attach the weight-tied head to the full final hidden-state sequence and train with shifted cross-entropy (language-modeling) loss on the training sonnets. Generation conditions on the first three lines of a held-out

sonnet and continues autoregressively with nucleus (top- p) sampling at $p = 0.9$ and temperature 1.2.

3.4 LoRA extension

For PEFT we add a low-rank update to each attention projection,

$$W'x = Wx + \frac{\alpha}{r} B(Ax), \quad A \in \mathbb{R}^{r \times d}, B \in \mathbb{R}^{d \times r}, \quad (6)$$

where W is frozen and only the low-rank factors A and B are trained. We place adapters on the query, key, and value projections of all 12 layers (36 adapter pairs) with rank $r = 8$ and $\alpha = 16$ (scaling 2). We compare three schemes for initializing A and B , contrasted in Figure 2. *default* draws A from $\mathcal{N}(0, 0.02^2)$ and sets $B = 0$, so the update is exactly zero at the start and is learned from scratch. *weight_dist* draws both factors from a Gaussian matched to the mean and standard deviation of the corresponding pretrained weight and scales them by 0.01, so the adapter begins with a small nonzero update in the numeric range of W . *svd* sets the factors from a truncated rank- r singular value decomposition of W (scaled by 0.01), so the adapter begins aligned with the dominant singular directions of the frozen weight.

3.5 Correctness validation

Before any training we confirm that our backbone reproduces the HuggingFace GPT-2 last hidden state to an absolute tolerance of 10^{-1} , and that our optimizer step matches a provided reference tensor to 10^{-6} . These checks isolate task results from implementation error.

4 Experiments

4.1 Datasets and statistics

Table 1 reports statistics computed directly from the provided splits, with sequence lengths measured in GPT-2 byte-pair tokens. The four tasks span very different input regimes: SST [10] sentences are short (≈ 22 tokens), CFIMDB reviews are an order of magnitude longer (≈ 218 tokens, up to the model context), Quora supplies a large but class-imbalanced corpus (63%/37%), and the sonnet corpus is tiny (131 training poems).

The backbone is GPT-2 small: 12 layers, hidden size 768, 12 heads, context length 1024, and a vocabulary of 50,257, totaling 124.44M parameters with the tied output head. The three adaptation regimes we compare differ sharply in how many of these parameters they train; we report each regime’s parameter budget next to its generation quality in Table 6.

4.2 Setup

Table 2 lists the training configuration of each task. All models use AdamW and the GPT-2 small backbone; sonnet decoding uses nucleus sampling. Sentiment and sonnet experiments were run on a single GPU; the Quora experiments used a larger batch on cloud GPU hardware.

4.3 Sentiment results

Table 3 compares our development accuracy and macro-F1 against the reference baselines from the assignment. On CFIMDB both regimes exceed the baseline, and full fine-tuning reaches 0.9837. On SST our accuracies are within half a point of the reference for both regimes. Figure 3 breaks SST down by class and exposes where the difficulty concentrates.

4.4 Paraphrase detection and the LoRA ablation

On the Quora development set the two-way head reaches 0.8978 accuracy and 0.8912 macro-F1 (Table 4). Per-class recall is balanced despite the 63%/37% label skew, indicating the classifier is not simply exploiting the majority class.

We run the same initialization ablation on paraphrase that we run on sonnets. Holding the configuration fixed (Table 2), we compare full fine-tuning against the three LoRA initializations on the development set (Table 5). Full fine-tuning reaches 0.896 accuracy in this controlled run.¹ The three LoRA initializations are essentially tied: they finish within 0.1 accuracy point of one another (last epoch 0.844–0.845, per-epoch peaks 0.844–0.849) and each recovers about 94% of full fine-tuning accuracy while training only 0.36% of the parameters. Figure 4 shows that the three adapters follow almost the same trajectory across all ten epochs. Unlike the sonnet ablation, where the weight-distribution initialization led, the choice of initialization is immaterial on this classification task.

4.5 Sonnet generation and the LoRA ablation

Full fine-tuning of the sonnet model reaches chrF 40.95 / BLEU 23.75 on the 12 held-out development poems.² Figure 5 shows the per-epoch trajectory: chrF rises steeply for the first six epochs and then plateaus, characteristic of a small dataset.

The ablation (Table 6) is the core experiment. Last-layer fine-tuning, despite training $16\times$ more parameters than LoRA, is the weakest adapter (88.2% of full chrF). All three LoRA variants surpass it while training only 0.36% of the model. The weight-distribution initialization is best, retaining 97.9% of full chrF; SVD and the default Gaussian initialization follow. Figure 6 visualizes this trade-off: the LoRA points sit on the upper-left Pareto frontier, achieving near-full quality at two orders of magnitude lower parameter cost.

4.6 Qualitative analysis

We categorize the generated sonnets into recurring behaviors (Table 9), using real outputs from the full model conditioned on held-out seeds. The model reliably reproduces the three-line prompt, adopts archaic diction, and preserves the indented closing-couplet cue, but it deviates from the strict 14-line iambic-pentameter form and occasionally emits coined or incoherent tokens.

5 Discussion and Conclusion

The gap between SST and Quora points to data quantity, rather than model capacity, as the limiting factor for full fine-tuning. SST provides only 8,544 short training sentences, and a 124M-parameter model fits them quickly without dev accuracy (0.5077) moving past the baseline. With 283k Quora pairs the same full-model recipe generalizes to 0.8978 accuracy and shows no comparable overfitting. On the smaller corpus, adding capacity is not what is missing.

¹The headline figure of 0.8978 in Table 4 comes from a separate run of the same full fine-tuning recipe; here we report the run that produced all three LoRA variants (0.896) so the four configurations share one training setup. The 0.0018 difference is run-to-run variation.

²An earlier independent run reached chrF 41.41 / BLEU 24.27; we report the run from which all ablation variants were produced so the comparison is controlled.

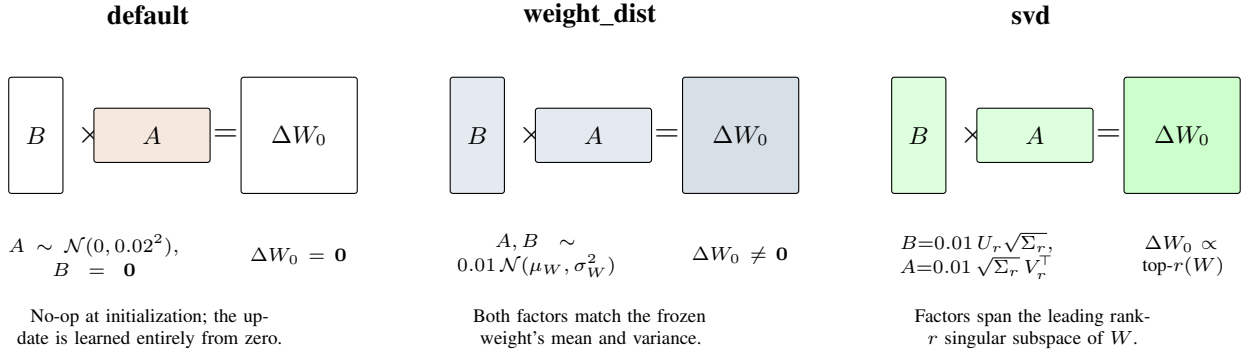


Figure 2: The three LoRA initialization schemes at the start of training. All use the same low-rank form $\Delta W = (\alpha/r) BA$ on each frozen Q/K/V projection ($r=8, \alpha=16$); they differ only in how the down-projection $A \in \mathbb{R}^{r \times d}$ (wide) and the up-projection $B \in \mathbb{R}^{d \times r}$ (tall) are initialized. *default* starts as a no-op ($B = 0$); *weight_dist* starts from a small Gaussian update matched to the frozen weight’s statistics; *svd* starts aligned with the frozen weight’s dominant singular directions. Here μ_W, σ_W are the mean and standard deviation of the pretrained projection W , and U_r, Σ_r, V_r its truncated rank- r singular value decomposition.

Dataset	Task	Cls.	Train	Dev	Test	Avg. len	Label balance
SST	5-way sentiment	5	8,544	1,101	2,210	22.4 tok	skewed (see note)
CFIMDB	binary sentiment	2	1,707	245	488	217.6 tok	$\approx 50/50$
Quora	paraphrase detection	2	283,011	40,429	80,860	27.0 tok	63%/37%
Sonnets	poem generation	n/a	131	12	12	160.0 tok	n/a

Table 1: Dataset statistics, computed from the provided files. Average length is in GPT-2 tokens: per sentence (SST, CFIMDB), per concatenated question pair (Quora), and per poem (Sonnets). SST training-class counts are $\{1092, 2218, 1624, 2322, 1288\}$ for the five ordinal classes; the neutral class is the minority among the central labels. The sonnet corpus contains 131 training poems (≈ 14 lines each) plus 12 held-out development and 12 held-out test poems.

Task	LR	Ep.	Batch	Drop.	Decode
Sentiment	10^{-5}	10	64/8	0.3	n/a
Paraphrase	$4e-5$	10	128	n/a	n/a
Sonnet	$1e-5$	10	8	n/a	$p=0.9, T=1.2$

Table 2: Training configuration. The learning rate shown for sentiment is the full fine-tuning value; the frozen-backbone probe uses the released default of 10^{-3} . Batch size is 64 (SST) and 8 (CFIMDB) with head dropout 0.3. LoRA runs use $r=8, \alpha=16$ on the Q/K/V projections of all 12 layers.

Data	Regime	Acc.	Mac-F1	Base.
SST	last-linear	0.4550	0.3788	0.462
	full-model	0.5077	0.4781	0.513
CFIMDB	last-linear	0.8776	0.8775	0.861
	full-model	0.9837	0.9837	0.976

Table 3: Sentiment dev results vs. assignment baselines. CFIMDB exceeds both baselines; SST is within 0.5 points.

The difficulty within a task is also uneven. SST’s neutral class is the hardest for both regimes (Figure 3: recall 0.183 for the full model and 0.079 for the probe), which fits its position between the ordinal extremes and its smaller share of the training labels. CFIMDB, a binary task with strongly polar reviews, is close to saturated at 0.9837.

Regularization interacts with the fine-tuning regime. The head dropout of 0.3 that helps the full model hurts the frozen-backbone probe: SST last-linear scores 0.4550 at dropout 0.3, whereas an earlier run without head dropout on the

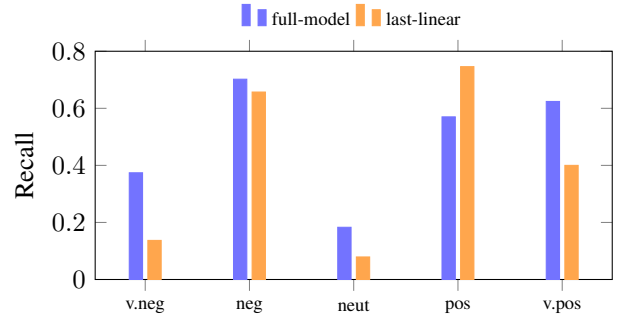


Figure 3: SST per-class dev recall. The neutral class is hardest for both regimes (full-model recall 0.183, last-linear 0.079); the frozen-backbone probe collapses on the rare extreme classes.

Metric	Value	Support
Accuracy	0.8978	40,429
Macro-F1	0.8912	n/a
Recall (non-dup., 0)	0.906	25,537
Recall (duplicate, 1)	0.884	14,892

Table 4: Quora paraphrase detection, development set.

probe reached about 0.476. A single global dropout setting is therefore not ideal, since the probe and the full model benefit from different values.

The ablation gives the clearest result. Spreading small low-rank updates across all layers (LoRA) works better than concentrating the same training budget in one layer: every LoRA variant outperforms last-layer fine-tuning while training $16\times$ fewer parameters. The weight-distribution initial-

Regime	Trainable	% tr.	Dev acc	Peak
Full fine-tune	124.44 M	100.00	0.896	0.896
LoRA (default)	0.44 M	0.36	0.845	0.848
LoRA (weight_dist)	0.44 M	0.36	0.844	<u>0.849</u>
LoRA (svd)	0.44 M	0.36	0.844	0.844

Table 5: Quora paraphrase LoRA ablation on the development set (40,429 pairs). “Dev acc” is the last-epoch checkpoint and “Peak” the best epoch. All runs share the configuration of Table 2 ($r=8$, $\alpha=16$, $lr=4e-5$, 10 epochs, batch 128) and the parameter budgets of Table 6; the LoRA budget adds a 1,538-parameter Linear(768, 2) classification head. The three initializations lie within 0.1 accuracy point of one another and each recovers about 94% of full fine-tuning accuracy.

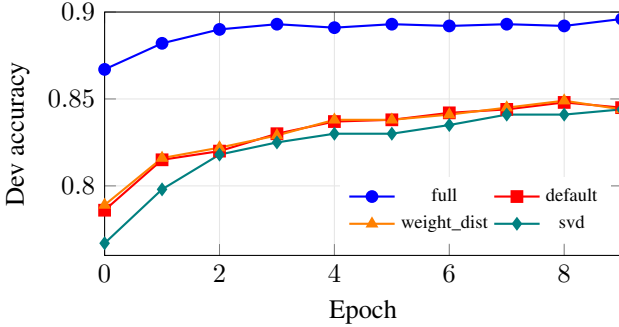


Figure 4: Quora development accuracy per epoch for the four paraphrase runs. The three LoRA initializations are nearly indistinguishable and settle about five accuracy points below full fine-tuning, in contrast to the sonnet ablation where the initializations separated.

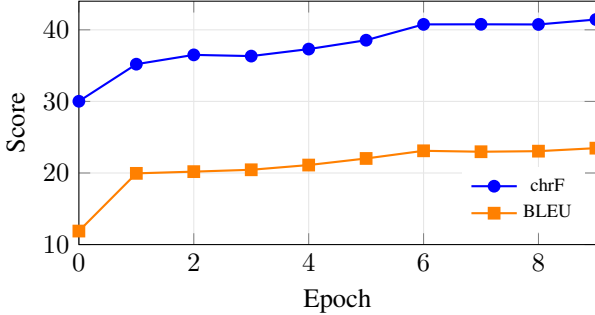


Figure 5: Full-model sonnet learning curve. chrF improves from 30.02 to 41.44 over ten epochs and plateaus after epoch six; BLEU tracks chrF at a lower absolute level.

ization, whose adapter begins in the geometry of the pre-trained weights (Figure 2), retains 97.9% of full chrF using 0.36% of the parameters and 1.8MB of storage.

The two ablations together sharpen this picture. Initialization is decisive only for generation: on sonnets the weight-distribution scheme retains 97.9% of full chrF while the default and SVD schemes retain 95.9% and 95.6% (Table 6), whereas on paraphrase the same three schemes finish within 0.1 accuracy point of one another and each recovers about 94% of full fine-tuning accuracy (Table 5). We attribute the difference to task conditions: the large Quora training set and a simple two-way head let any near-zero-perturbation start converge to the same solution, whereas low-resource sonnet generation is more sensitive to where in weight space

Regime	Trainable	% tr.	chrF	BLEU
Full fine-tune	124.44 M	100.00	40.95	23.75
Last-layer	7.09 M	5.70	36.11	21.58
LoRA (default)	0.44 M	0.36	39.29	23.07
LoRA (weight_dist)	0.44 M	0.36	<u>40.09</u>	<u>23.54</u>
LoRA (svd)	0.44 M	0.36	39.13	23.10

Table 6: Adaptation regimes on the held-out sonnets: trainable-parameter cost against generation quality. “% tr.” is the fraction of the 124.44M model that is trained. LoRA trains $281\times$ fewer parameters than full fine-tuning (a 1.8MB adapter versus a 498 MB checkpoint), yet the weight-distribution initialization retains 97.9% of full chrF; last-layer fine-tuning trains $16\times$ more parameters than LoRA but reaches only 88.2%. Storage is derived at 4 bytes/parameter.

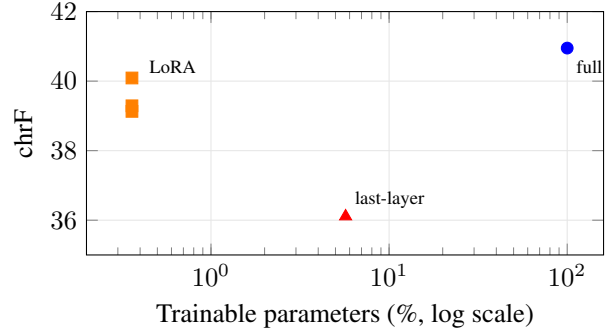


Figure 6: Quality versus trainable-parameter cost. The three LoRA variants (orange) reach near full fine-tuning chrF (blue) at 0.36% trainable parameters, dominating last-layer fine-tuning (red).

adaptation begins.

For the sonnet task the choice of metric matters. BLEU stays below chrF throughout because, on poems of roughly 14 lines, word n -gram precision is held down by the brevity penalty and by sparse matches, while the character-level chrF tracks adherence to Shakespearean form more closely. We therefore report chrF as the primary score.

A few limitations remain. We report development metrics only, since the test labels are withheld; the paraphrase train and test prompts use different templates, so the dev figure may be optimistic for the test distribution; and each configuration was run once, so we do not measure variance. Natural extensions include preference optimization for generation quality, applying LoRA to larger GPT-2 variants, and testing alternative attention mechanisms.

We implemented and validated GPT-2 and AdamW ourselves, adapted the model to three tasks with quantitative results, and found that a carefully initialized LoRA adapter recovers almost all of full fine-tuning’s generation quality at a small fraction of its parameter and storage cost.

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A Per-Epoch Sonnet Metrics (Full Model)

Table 7 lists the complete per-epoch training loss and dev metrics behind Figure 5.

Epoch	0	1	2	3	4	5	6	7	8	9	Final
Train loss	4.799	4.437	4.279	4.181	4.096	4.047	4.000	3.949	3.916	3.877	n/a
chrF	30.02	35.20	36.50	36.33	37.31	38.55	40.76	40.77	40.75	41.44	40.95
BLEU	11.89	19.95	20.19	20.45	21.11	22.03	23.10	22.97	23.05	23.47	23.75

Table 7: Full-model sonnet training: per-epoch loss and held-out chrF/BLEU. “Final” is the score of the last-epoch model used for submission.

B Per-Epoch Paraphrase Development Accuracy

Table 8 lists the complete per-epoch Quora development accuracy behind Figure 4 for the four paraphrase runs.

Epoch	0	1	2	3	4	5	6	7	8	9
Full fine-tune	0.867	0.882	0.890	0.893	0.891	0.893	0.892	0.893	0.892	0.896
LoRA (default)	0.786	0.815	0.820	0.830	0.837	0.838	0.842	0.844	0.848	0.845
LoRA (weight_dist)	0.789	0.816	0.822	0.829	0.838	0.838	0.841	0.845	0.849	0.844
LoRA (svd)	0.767	0.798	0.818	0.825	0.830	0.830	0.835	0.841	0.841	0.844

Table 8: Per-epoch Quora development accuracy for the four paraphrase runs (single seed). The epoch-9 column corresponds to the “Dev acc” values in Table 5.

C Qualitative Taxonomy of Sonnet Generation

Category	Description	Example (real generated output)
Seed fidelity	The three conditioning lines are reproduced verbatim.	“Thine eyes I love, and they, as pitying me, / Knowing thy heart torment me with disdain,”
Archaic diction	Period-appropriate vocabulary and contractions are adopted.	“Whate’er I have gathered...”, “Lo, however... ”
Structural cue	The indented closing-couplet layout is preserved.	“ Lo, however no dirty day glory will blame thee...”
Length/meter deviation	Output departs from the strict 14-line iambic-pentameter form.	Generated stanzas vary in line count and rhythm.
Lexical artifacts	Coined or non-words appear.	“desexor”, “priggish pompous verse”
Semantic drift	Thematic coherence degrades over the stanza.	Later lines lose connection to the opening conceit.

Table 9: Taxonomy of generation behaviors observed in full-model sonnets, with verbatim examples.